

HR-FONT: COMPLETING FONT FAMILIES ACROSS SCRIPTS

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ABSTRACT

We introduce HR-Font, a framework for extending a font family to new scripts from a few Chinese reference glyphs. The central challenge is to translate a family’s visual language into characters whose structures differ from those observed. HR-Font addresses this challenge through two complementary forms of conditioning. Dynamic Delta expresses same-character examples from a font bank as anchor-relative structural variations and learns to combine them as generation unfolds. Target-character appearance completion reads multi-scale local reference features to predict a spatial appearance representation of the requested glyph, supervised by training-target features. The resulting conditions guide a diffusion generator toward recognizable characters with a coherent family identity. We also formulate a cross-script completion benchmark that evaluates character identity, reference consistency, and family compatibility, with script-aware data curation and controlled reference budgets.

1 INTRODUCTION

A designer extending a Chinese typeface to Latin or kana needs more than a recognizable outline. The new glyphs should belong to the same family: weight, contrast, terminals, and decorative treatment should feel coherent across scripts. A few Chinese examples provide evidence of that visual language, but they leave its realization in the target characters open. The design of a Latin *g*, for example, requires choices that are not directly visible in a Chinese reference glyph. Cross-script font completion therefore calls for both a vocabulary of target-character variations and a model of how the observed appearance applies to each character.

Few-shot generators learn to synthesize glyphs from content and reference images. FontDiffuser uses conditional diffusion and multi-scale content aggregation (Yang et al., 2023); FSFont learns fine-grained correspondence between content and reference features (Tang et al., 2022). Font-bank methods such as CF-Font enrich content representations using corresponding glyphs from basis fonts (Wang et al., 2023). These approaches provide useful ingredients for completion. The cross-script setting makes a further distinction important: available glyph variations describe possible designs, whereas the reference images specify the family to which the result should belong.

We develop HR-Font around this distinction. The first component, *Dynamic Delta*, represents a requested character through anchor-relative feature changes supported by a font bank. The neutral rendering supplies identity, while a learned router combines the available variations at each spatial location and denoising step. The second component, *target-character appearance completion* (TC), predicts a spatial appearance representation for the requested character from its neutral content and the observed references. Its online reference reader preserves local evidence before organizing that evidence around the target character. Training glyphs provide direct supervision for this prediction. Together, these components connect *how the character can vary* with *how the family’s appearance should be realized*. Figure 1 summarizes the model.

This formulation also shapes the evaluation. Matching a held-out target image is useful, but a plausible completion can differ from that particular design. We therefore assess three outcomes: whether the glyph has the requested identity, whether it expresses the references’ visible attributes, and whether it fits the target family across scripts. Script-aware curation establishes which font–character pairs provide meaningful supervision, while fixed reference episodes make comparisons reproducible.

Inputs	Conditions for the generator
Neutral target glyph	$x_c^0 \rightarrow E_c(x_c^0)$ Character identity
Same-character bank + reference retrieval	$d_j = E_c(B_{j,c}) - E_c(x_c^0)$ $\Delta_{\ell,t,p} = \sum_j w_{j,\ell,t,p} d_{j,p}$ Dynamic Delta: adaptive structural variations
Reference pixels + target-content grid	$720n$ local tokens $\rightarrow$ 144 target-aligned slots TC memory $\rightarrow$ appearance tokens + feature prediction Training target $\rightarrow$ spatial VGG teacher
Synthesis	Global attention + gated target-appearance attention Neutral content + Dynamic Delta + TC memory $\rightarrow$ diffusion denoising $\rightarrow \hat{y}_{f,c}$

Figure 1: Two complementary conditions for cross-script completion. Dynamic Delta selects and combines anchor-relative structural variations from a font bank. TC reads local reference evidence into a target-aligned appearance memory. A diffusion generator uses both conditions alongside neutral content and global reference features. Training-target features supervise the appearance memory.

Our contributions are:

- A cross-script font completion formulation and evaluation protocol that distinguish character identity, reference consistency, and family compatibility.
- Dynamic Delta, an anchor-relative glyph variation space with spatially and temporally adaptive combination of corresponding training-bank examples.
- Target-character appearance completion, an online local-reference reader that learns a target-aligned appearance memory through direct feature supervision.

2 RELATED WORK

Cross-script and few-shot font generation. FTransGAN studies few-shot transfer between languages with multi-level attention (Li et al., 2021). FontDiffuser generates glyphs by conditional denoising and uses multi-scale content aggregation and style contrastive learning (Yang et al., 2023). FSFont uses content queries to aggregate fine-grained local reference styles (Tang et al., 2022). HR-Font builds on conditional generation and local reference evidence, and introduces a supervised intermediate prediction: the appearance of the requested target glyph. This prediction connects reference observations to the target character before the generator renders the image.

Font-bank content representations. CF-Font projects content features into a space spanned by corresponding glyphs from basis fonts (Wang et al., 2023). Dynamic Delta shares the use of same-character bank evidence. Its interface expresses that evidence as a residual from a fixed neutral anchor and consumes the residual through the structural conditioning path, alongside separately supplied neutral content. The shared anchor keeps the content input stable while a denoising-conditioned router adapts the mixture of residuals. Our evaluation tests this interface under shared bank access and training budgets.

Appearance transfer and completion. Feature statistics provide appearance representations in neural style transfer (Huang & Belongie, 2017), while local attention retrieves features from observed glyphs (Tang et al., 2022). TC organizes such observations into a target-aligned spatial memory. Its distinctive learning objective predicts the feature map of a different character in the same family. This supervision shapes both reference extraction and target-conditioned aggregation, and the resulting memory enters the generator through an additional reference-attention readout.

3 CROSS-SCRIPT FONT COMPLETION

Let c be a requested target character and x_c^0 its rendering in a neutral font. For a target family f , the observed reference set is $\mathcal{R}_f = \{(a_i, r_i)\}_{i=1}^n$, where a_i is a Chinese character and r_i its rendering in f . The task is to generate $\hat{y}_{f,c}$ with the identity of c and an appearance compatible with f . The evaluation font is held out from training. A bank $\mathcal{B}$ contains training-font renderings of the reference and target characters. During training, an episode excludes its own font from donor retrieval.

Three evaluation dimensions. *Identity* measures whether the output represents the requested character. *Reference consistency* measures agreement with visible attributes such as weight, contrast, stroke endings, and texture. *Family compatibility* measures whether the generated glyph is a coherent extension of the family across scripts. The latter includes character-specific decisions whose exact realization is not shown in the reference set. Paired target-image measurements complement these dimensions with reconstruction diagnostics.

Script-aware supervision. A font file can contain several scripts with different design conventions, fallback glyphs, or inconsistent coverage. We curate valid font-character pairs and eligible donors by script, preserving the font-level train/validation/test split. A mismatch in design conventions calls for family-level inspection; automatic similarity alone does not determine validity. The released protocol records pair validity, donor eligibility, and the reference episode used for each output.

4 METHOD

4.1 OVERVIEW

We use a conditional diffusion generator with a frozen content encoder E_c and a frozen global style encoder E_s . The content encoder supplies multi-scale features of the neutral target; the global style encoder supplies reference tokens and bank-retrieval descriptors. A separate trainable local encoder E_ℓ , initialized from the early blocks of E_s , reads the reference pixels online. Dynamic Delta conditions structural interaction, while TC turns local evidence into target-aligned appearance tokens. The denoiser thus receives

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, E_c(x_c^0), \Delta_{c,t}(\mathcal{R}_f), G(\mathcal{R}_f), A(\mathcal{R}_f, c)). \quad (1)$$

Here x_t is the noisy target image, G contains nine global reference tokens, and A contains 144 appearance tokens arranged around the target’s 12×12 content grid.

4.2 DYNAMIC DELTA: A STRUCTURAL VARIATION SPACE

The bank supplies examples of how the requested character is realized in different fonts. To prioritize a relevant neighborhood, we compare the target references with bank renderings of the same reference characters:

$$s_j(\mathcal{R}_f) = \frac{1}{n} \sum_{i=1}^n \cos(\bar{E}_s(r_i), \bar{E}_s(B_{j,a_i})), \quad (2)$$

$$\alpha_j = \frac{\exp(s_j/\tau)}{\sum_{\ell \in \mathcal{J}_c(\mathcal{R}_f)} \exp(s_\ell/\tau)}, \quad j \in \mathcal{J}_c(\mathcal{R}_f). \quad (3)$$

$\bar{E}_s$ is a pooled descriptor. $\mathcal{J}_c$ contains the top- K eligible training fonts after removing the episode font and script-invalid donors. The implementation uses $K = 10$ and $\tau = 0.07$.

For each content scale s , we retain a set of anchor-relative candidates:

$$d_{j,c}^s = E_c^s(B_{j,c}) - E_c^s(x_c^0), \quad j \in \mathcal{J}_c. \quad (4)$$

Every donor renders the same requested character. Subtracting a common anchor expresses the bank evidence in coordinates relative to the content already available to the generator. These candidates define a bank-supported space of possible changes to the neutral structure. Retrieval supplies an

initial distribution α , and a learned router adapts that distribution to the current synthesis state. At location p of structural module ℓ , the router computes

$$q_{\ell,t,p} = Q_{\ell}(h_{\ell,t})_p + C_{\ell}(E_c^s(x_c^0))_p + R_{\ell}(\bar{G}) + T_{\ell}(t), \quad (5)$$

$$w_{j,\ell,t,p} = \text{softmax}_j [\log \alpha_j + \gamma_{\ell} \cos(q_{\ell,t,p}, K_{\ell}(d_{j,c}^s)_p)], \quad (6)$$

$$\Delta_{c,\ell,t,p} = \sum_{j \in \mathcal{J}_c} w_{j,\ell,t,p} d_{j,c}^s. \quad (7)$$

$h_{\ell,t}$ is the generator’s content-side hidden state, $\bar{G}$ is the mean global reference token, and γ_{ℓ} is learned. The resulting residual drives deformable structural interaction with the neutral content.

The weights are nonnegative and sum to one: adaptation selects within the available residual variations at each location. Reference similarity narrows the search neighborhood; target content and the evolving denoising state determine how it is used. This separates neighborhood retrieval from cross-script realization. Mean-Delta is recovered by fixing $w_j = \alpha_j$, providing a direct comparison for learned combination.

4.3 TARGET-CHARACTER APPEARANCE COMPLETION

TC learns how local reference observations should be expressed in the requested character. It first extracts evidence at multiple scales, then arranges that evidence into a target-conditioned memory shared by appearance prediction and synthesis.

Online local evidence. For each reference, E_{ℓ} produces a $24 \times 24 \times 128$ feature map and a $12 \times 12 \times 256$ map. A 1×1 projection maps the shallower features to 256 channels. Flattening and concatenating both maps yields 720 tokens per reference. All valid references form a joint evidence sequence $V_f \in \mathbb{R}^{720n \times 256}$. The encoder processes valid images only, and attention masks padded positions. Thus, observations remain available individually until the target-conditioned readout.

Target-aligned memory. The neutral content grid $U_c \in \mathbb{R}^{144 \times 256}$ defines one query per target location. Learned positional slots P complement this content signal:

$$Q_c = W_q U_c + P, \quad (8)$$

$$H_{f,c} = \text{Attn}(Q_c, W_k V_f, W_v V_f), \quad (9)$$

$$M_{f,c} = H_{f,c} + \text{MLP}(\text{LN}(H_{f,c})), \quad (10)$$

$$A_{f,c} = W_o M_{f,c}, \quad \hat{F}_{f,c} = W_a M_{f,c}. \quad (11)$$

Attention uses four heads, per-head query/key normalization, and FP32 softmax. $A_{f,c} \in \mathbb{R}^{144 \times 1024}$ conditions synthesis, while $\hat{F}_{f,c} \in \mathbb{R}^{144 \times 128}$ is the supervised appearance readout. The target queries can select different local evidence even when only one reference glyph is available.

Supervision and injection. For training target $y_{f,c}$, a frozen VGG16 (Simonyan & Zisserman, 2015) feature extractor supplies $F_{f,c}^* = S(\text{Pool}_{12 \times 12}(\phi_2(y_{f,c})))$. The second feature map has 128 channels; S uses per-channel statistics fitted on clean training targets. Retaining the spatial grid supervises where target-specific appearance should be expressed. TC minimizes

$$\mathcal{L}_{\text{comp}} = \text{SmoothL1}(\hat{F}_{f,c}, \text{stopgrad}(F_{f,c}^*)). \quad (12)$$

At each up-path reference-attention layer, the generator reads the global tokens and the appearance memory separately:

$$Z = \text{CA}(h, G) + \eta r(u) \text{CA}(h, A_{f,c}). \quad (13)$$

The two reads share attention projections. A learned gain η , initialized at 0.01, and a 1,000-update ramp $r(u)$ introduce the new appearance path gradually. The global tokens summarize family context, while the spatial memory supplies target-specific local evidence. The supervised readout and the synthesis tokens share $M_{f,c}$, so both objectives shape the online reference encoder and its aggregation.

4.4 TRAINING AND GENERATION

We jointly optimize the local reader, Delta router, and denoiser from a pretrained generation checkpoint:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + 0.01\mathcal{L}_{\text{perc}} + 0.25\mathcal{L}_{\text{off}} + 0.01r(u)\mathcal{L}_{\text{comp}}. \quad (14)$$

$\mathcal{L}_{\text{diff}}$ is the epsilon-prediction mean squared error; The perceptual term averages three VGG feature losses; $\mathcal{L}_{\text{off}}$ is the raw offset regularizer returned by the backbone. The content, global style, and VGG encoders stay frozen; the local reference encoder is trainable. Each training episode uses one to eight Chinese references and a valid target glyph from the same font.

The TC teacher is used only for its supervised loss. The denoiser always receives the predicted appearance memory, including during training. Classifier-free condition dropout jointly removes the global reference, local appearance, neutral-content, and structural conditions from the denoiser. The completion loss retains its observed inputs; an additional source-drop event removes structural offsets. At inference, TC is evaluated once for each content-reference pair and reused across denoising steps, while Delta weights adapt at every structural module and step.

Within-script auxiliary supervision. We also study same-family Chinese completion as an auxiliary training task. A Chinese target is sampled from an eligible training font, and its character is excluded from the reference set. The auxiliary task shares the reader and denoiser, uses neutral Chinese content, and supervises appearance with the same objective. Its structural-bank offsets are disabled, keeping the auxiliary signal focused on reference-based completion. The combined objective is $\mathcal{L}_{\text{main}} + 0.5\mathcal{L}_{\text{CN}}$. Cross-script validation determines whether this additional supervision improves the main task.

5 EXPERIMENTS

5.1 DATA AND SETTINGS

The corpus contains 260 fonts partitioned into 228 training, 16 validation, and 16 test fonts. The inventory comprises 295 target characters and 338 Chinese reference characters. Protocol A renders RGB glyphs at 96×96 , with a shared font size within each font and neutral Noto Sans CJK Regular content. The current clean training mapping retains 56,429 target pairs. The stratified evaluation panel spans 47 characters covering Latin, diacritics, kana, and phonetic symbols; validity filtering retains 704 test pairs.

Reference-budget evaluation uses nested sets of 1, 2, 4, and 8 characters. All methods in a comparison receive the same ordered reference set, and Delta retrieval uses that episode’s references. The main reference manifest and checkpoint selection are finalized on validation data. The appendix records reference characters, bank eligibility, sampler, and provenance.

Training settings. The current joint experiments use eight V100 GPUs, eight samples per device, FP16 arithmetic, and gradient clipping at 1.0. The effective main-task batch size is 64. The denoiser and inherited structural modules use a peak learning rate of 2×10^{-5} ; the local reader and router use 10^{-4} . Joint training runs for 10,000 successful updates, with 500 warmup updates, a plateau through update 5,000, and cosine decay to 10% of the peak rate. The auxiliary experiment adds 32 Chinese examples per update and starts independently from the same parent weights. EMA weights are evaluated every 2,000 updates. Complete resolved settings appear in Appendix A.

5.2 COMPARISONS AND MEASUREMENTS

External comparisons cover conditional diffusion (FontDiffuser), cross-language transfer (FTransGAN), font-bank content fusion (CF-Font), and fine-grained reference attention (FSFont). Each method is accompanied by its reference budget, bank access, pretraining data, and cross-script adaptation. Internal comparisons isolate dynamic structural combination, local appearance completion, and auxiliary supervision using shared initial weights and data protocols. The strongest validation baseline is retained in the main comparison.

Table 1: Main comparison under the cross-script protocol. Identity and style preference are percentages; family compatibility uses the cross-script cosine score. Empty cells await completed evaluation.

Method	Identity $\uparrow$	Style pref. $\uparrow$	Family $\uparrow$	LPIPS $\downarrow$
FontDiffuser	–	–	–	–
FTransGAN	–	–	–	–
CF-Font	–	–	–	–
FSFont	–	–	–	–
Mean-Delta baseline	–	–	–	–
HR-Font	–	–	–	–

Table 2: Contribution studies. The execution plan prioritizes the full model and auxiliary study, followed by targeted shared-budget comparisons.

Question	Matched comparison	Result
Structural source	Bank-supported Delta versus reference-derived structure	–
Residual interface	Anchor-relative versus absolute bank features	–
Dynamic combination	Routed versus fixed Mean-Delta, same donors	–
Appearance supervision	Spatial teacher versus generation loss only	–
Character conditioning	Content-plus-slot versus slot-only queries	–
Reference adaptation	Trainable versus frozen local encoder	–
Complementarity	Shared-parent structural / appearance / full model	–
Auxiliary supervision	Full model with versus without Chinese completion	–

We report character recognition accuracy, blind paired style preference, and family compatibility. Human comparison displays the neutral target and the same reference set with randomized method positions and an explicit tie option. The learned compatibility score uses a separately trained cross-script encoder and cosine similarity between a generated glyph and a Chinese-reference prototype. We distinguish its font-retrieval validation from agreement with human judgments of generated appearance. L1, SSIM, and LPIPS provide paired reconstruction diagnostics. Results are aggregated by font and script over the same valid examples.

5.3 CONTRIBUTION STUDIES

Glyph variation prior. We compare dynamic combination with fixed Mean-Delta weights over the same retrieved candidates. This comparison preserves bank access, anchor-relative features, and the appearance interface. A same-bank absolute-feature control tests anchor-relative conditioning, while a shared-schedule structural-source comparison establishes the benefit of bank-supported variations.

Target-character appearance completion. We compare the full reader with a same-capacity branch trained using the generation objective alone. A learned-slot-only query tests the role of target content, and a frozen-local-encoder variant tests adaptation of reference extraction. These comparisons retain the 144-token interface, reference pixels, and training schedule.

Complementarity. Shared-parent variants separately add the structural and appearance components, followed by the full model. The Chinese auxiliary study then measures whether additional same-family completion improves cross-script synthesis. All variants are evaluated with identical reference episodes and diffusion noise. Inference-time condition removal provides a separate sensitivity analysis of the trained model.

5.4 REFERENCE BUDGETS AND QUALITATIVE ANALYSIS

Reference-budget curves evaluate the same checkpoints at 1, 2, 4, and 8 shots. They measure how additional observations affect identity, appearance, and family coherence. Qualitative comparisons include ordinary stroke styles, decorative fonts, and difficult cross-script realizations. For each selected row, all methods share the target, references, sampler, and noise. Figure 2 reserves the comparison layout for the completed outputs.

Case	References	Baseline	Mean-Delta	HR-Font	Target
Stroke style					
Decoration					
Difficult completion					

Figure 2: Qualitative comparison layout. Rows cover stroke style, decorative treatment, and difficult cross-script completion. Blank cells will contain matched generated glyphs and the corresponding target design.

6 DISCUSSION

The two conditions provide different forms of information. Dynamic Delta brings target-character examples into the generator’s content coordinates and adapts their combination during synthesis. TC learns a target-specific spatial appearance prediction from observed references. Their combination gives the generator a structural prior and a supervised appearance destination. The contribution studies measure how these roles translate into image quality.

The current representation operates on 96×96 raster glyphs. Spatial appearance supervision connects local reference evidence to target locations. The denoiser integrates this evidence into complete strokes, terminals, and decorative patterns. Bank coverage also determines the available examples for rare target designs. Extending the formulation to higher-resolution glyphs and complete typographic layout is a natural next step.

7 CONCLUSION

HR-Font frames cross-script font generation as coherent family completion. Dynamic Delta provides an adaptive bank-supported variation prior relative to neutral content, and target-character appearance completion predicts how local reference observations apply to the requested glyph. The accompanying evaluation separates identity, reference consistency, and family compatibility. Together, the formulation and two conditioning components establish a practical framework for extending a font’s visual language across scripts.

REPRODUCIBILITY STATEMENT

The supplement records data mappings, cache fingerprints, reference episodes, parent checkpoints, resolved training configurations, and evaluation scripts. The paper’s editable figure sources accompany the draft. Final results will include measured training and inference costs.

ETHICS STATEMENT

Font completion can reduce repetitive design work and improve script coverage. Font licenses and source provenance must govern research use and redistribution. The release will document permitted assets and the consent, compensation, and review procedures for human evaluation.

AI USE STATEMENT

AI tools assisted with research discussion, code review, drafting, and figure layout. Experimental measurements come from the recorded training and evaluation runs. The authors review the resulting text and artifacts and take responsibility for the submission.

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A REPRODUCIBILITY DETAILS

Table 3: Current implementation settings. Final run hashes and measured costs are filled from the selected artifacts.

Setting	Value
Image resolution	96 × 96 RGB, Protocol A
Font split	228 / 16 / 16
Valid training target pairs	56,429
Delta retrieval	top-10, temperature 0.07
Delta router	width 32; location/module/time-conditioned weights
TC local evidence	$24^2 + 12^2 = 720$ tokens per reference
TC memory	144 target queries, width 256, four attention heads
TC output / teacher	$144 \times 1024 / 144 \times 128$
Reference injection	additive up-path attention; gain initialized at 0.01
Optimizer	AdamW, (0.9, 0.999), $\epsilon = 10^{-8}$, weight decay 0.01
Joint batch / precision	$8 \times 8 = 64 / \text{FP16}$
Joint learning rates	inherited 2×10^{-5} ; new 10^{-4}
Schedule	500 warmup; plateau to 5k; cosine decay to 10k
Gradient clipping	1.0
Loss coefficients	perceptual 0.01; raw offset 0.25; completion 0.01
Auxiliary Chinese study	batch 32; weight 0.5; shared main batch 64
Reference / source dropout	0.10 / 0.25
Parent checkpoint hashes	–
Mapping and cache hashes	–
Training GPU hours / peak memory	–
Inference latency	–

Implementation details. VGG inputs use ImageNet normalization; the teacher is adaptively pooled to 12×12 and standardized using clean training-target channel statistics. Joint stages load the same parent checkpoint and create a new optimizer and scheduler. The local encoder is a trainable copy of the first three global-style encoder blocks. Reference attention uses normalized 64-dimensional query/key heads with FP32 logits and softmax. Delta uses 32-dimensional routing features, with 1×1 hidden/content projections and a 3×3 candidate projection. Its time embedding maps $[t/1000, (t/1000)^2]$ through a two-layer MLP; the routing strength starts at 0.1. EMA decay rises to a maximum of 0.999, and the completion coefficient and local attention contribution ramp over 1,000 updates.

Run correspondence. The I0 reference is the 10,000-update G0b V0913 checkpoint. I1 trains the full method; I2 adds Chinese auxiliary supervision and initializes independently from I0. The Chinese auxiliary neutral inventory contains 338 glyphs rendered with the Protocol-A rule at font size 83; its images and frozen-content features have a separate manifest. Chinese targets and references reuse the original training PNGs, and target characters are excluded from their reference sets.

Randomness and uncertainty. Training uses random state 3407. Evaluation pairs reference episodes and diffusion noise across methods. Planned 95% intervals resample fonts while keeping paired method outputs together. These intervals describe variation across evaluated fonts.

Reference episodes and sampler. The primary nested episode is specified by Unicode code points: U+6C38, U+548C, U+4E66, U+98CE, U+9AA8, U+97F5, U+5929, U+5730. The 1/2/4/8-shot sets use prefixes of this list. The periodic validation panel uses 192 outputs at 1/4/8 shots. The scheduled final validation panel comprises 16 validation fonts, 16 valid target characters per font, four reference episodes, and four budgets, yielding 4,096 outputs. Inference uses DPM-Solver++ order 2, 20 steps, and guidance scale 7.5. The larger 47-character protocol remains a separate evaluation panel with its own valid-pair manifest.

Model selection. Validation panels guide model and checkpoint selection using appearance, readability, and identity. The final selection uses the full clean validation split. Exploratory test visualizations are retained with their dates; subsequent selection uses validation data. All reported methods are scored over an explicitly intersected valid sample set.

B CROSS-SCRIPT COMPATIBILITY EVALUATION

The current E12-b encoder ϕ is a ResNet-18 trained from random initialization on paired Chinese and Latin ground-truth glyphs. It maps each 96×96 image to a unit-normalized 512-dimensional feature. Training uses symmetric cross-script InfoNCE at temperature 0.07. The scorer’s data are drawn from 223 usable fonts in the generator’s training partition; the generator’s validation and test font files are excluded from scorer fitting. The internal scorer split contains 156/33/34 font files for training, validation, and testing. For a scoring reference set $\mathcal{R}_f$, define

$$p_f = \frac{\sum_{r \in \mathcal{R}_f} \phi(r)}{\|\sum_{r \in \mathcal{R}_f} \phi(r)\|_2}, \quad S(\hat{y}, \mathcal{R}_f) = \frac{1 + \langle \phi(\hat{y}), p_f \rangle}{2}. \quad (15)$$

The score evaluates compatibility with the supplied references and requires no target image at scoring time. Its retrieval study ranks Chinese reference prototypes using Latin glyph queries. Its verification study compares same-font and different-font pairs. The primary encoder study covers 52 Latin letters; script-specific validation accompanies extensions of this score to other target scripts.

A separate membership head combines query features with mean/max-pooled reference features and predicts same-font membership. Its temperature is fitted on internal validation data; we treat this probability as a supplementary diagnostic. Human evaluation measures visual agreement on generated outputs through blinded, paired judgments and font-clustered uncertainty. Numerical evaluator validation and generation results will be added after the current protocol review and paired evaluation are finalized.

C ADDITIONAL RESULTS

Table 4: Reference-budget evaluation of the selected HR-Font checkpoint.

References	Identity $\uparrow$	Style pref. $\uparrow$	Family $\uparrow$	LPIPS $\downarrow$
1	—	—	—	—
2	—	—	—	—
4	—	—	—	—
8	—	—	—	—